

Ordo Take-Home — A Vision Model on a Phone

Submission report · Anil Sahith · OnePlus 15R (Snapdragon SM8845, 12 GB) · champion: Qwen3-VL-2B Q4_0 · llama.cpp + Adreno GPU *Living document* — *regenerated as pending runs land*. Companion detail: [LAB NOTES.md](#) (every measurement) · [JOURNEY.md](#) (decision log) · [QA.md](#) · [BEST RESULT.md](#) · live dashboard (`dashboard/`) with per-run drill-down.

Provenance. Every inference in this project — every model, runtime, backend, token budget, on both devices — is one row in an append-only ledger, exported chronologically to [results/history_all_runs.csv](#) (regenerated with every render of this document). Each row carries the full history columns: timestamp, device/runtime/engine, variant, token cap, measured prompt tokens, encoder/decoder placement with its evidence, TTFT/encode/prefill/decode timings, peak RAM & SoC temperature, the exact question, the model's raw answer, the ground truth it was scored against, and the verdict. Every table in this report is an aggregation of those rows; nothing was measured off-ledger.

Part 1 — Getting a vision model onto a phone

Model choice. Started with Qwen2.5-VL-3B-Instruct (best OCR-class reader in the 0.3–4 B range, official multi-precision GGUFs, first-class llama.cpp support, dynamic-resolution ViT = a controllable image-token knob). Mid-project, measurement dethroned it: **Qwen3-VL-2B-Instruct** equals or beats it on accuracy while being ~2× faster at every stage and half the download (1.0 GB). Final bracket measured: Qwen3-2B, Qwen2.5-3B, SmolVLM-500M, SmolVLM2-2.2B, LFM2-VL-450M & 1.6B.

How it runs. llama.cpp cross-compiled with the Android NDK (arm64, i8mm/KleidiAI), binaries in `/data/local/tmp` over **wireless adb** — no app, no root, driven from a laptop harness + live dashboard. Three builds: Adreno-OpenCL (champion), tuned-CPU, and a **portable** `armv8.2-a` **compat build** that runs on any recent arm64 Android at only ~20% prefill cost.

What had to be patched. llama-server exposes no per-request vision-encode timing — and stage separation is the heart of this assignment — so we patched `tools/server/server-context.cpp` to log it. (MNN needed `MNN_BUILD_OPENCV=ON` or it *silently ignores images* — trap #7 below. LiteRT needed a hand-built stub library exporting three no-op symbols to satisfy an undistributed dependency.)

Runtimes tried (full grid in Part 3): **llama.cpp** (winner), **MNN** (own build: text engine excellent, best cold-start at 7.5 s incl. load; vision encoder 5–6× slower on both platforms), **LiteRT** (fastest model init measured, 0.8 s for 3 GB; slowest compute; vision locked inside app SDKs; macOS binaries unbuildable without full Xcode), plus documented assessments of GenieX (device-gated), MediaPipe/Gallery (per-SoC APKs carrying a QNN stack).

GPU or NPU? (Full per-model stage→backend map in Part 3.) **GPU: yes — both stages, proven by ablation:** encoder 7.8 s on Adreno vs 22.6 s forced-CPU (2.9×); decoder prefill ~410 vs ~55 t/s (~7×).

NPU: no accessible path on this SoC, evidenced at every layer — no Hexagon HTP runtime libs in `/vendor/lib64`; LiteRT accepts `--backend=npu`, loads the QNN libs, then the public model bundle itself

refuses (“model requires [cpu,gpu]” — NPU needs per-SoC compiled bundles; only a MediaTek one exists publicly). On Snapdragon today the NPU is vendor-privileged.

Shipping a 1–2 GB model to users. Not in the APK: post-install CDN download (WorkManager: resumable, checksummed, Wi-Fi-gated) or Play Asset Delivery (2 GB packs); iOS: On-Demand Resources. Quantization is a distribution feature — our champion is a 1.0 GB download vs 5.8 GB F16.

Seven measurement traps found and fixed en route (each silently corrupted numbers): prompt-cache faking repeat TTFTs · encode hidden inside “prefill” · sentence-vs-span ANLS scoring · verbose models truncated before their answer · constant-95 °C trip-point pseudo-sensors · a battery-charge zone masquerading as 57 °C thermal · a build flag that silently drops image input.

Part 2 — The eval set (camera-shot upgrade pending)

Construction. Three tiers: (1) a 24-image **synthetic dev set** (rendered menus/receipts/medicine/signs/handwriting/whiteboards/spines/appliance displays with controlled rotation/blur/dim/glare/occlusion; exact auto-generated ground truth; 9/9/6 easy/medium/hard) used for pipeline mechanics; (2) the **30-photo real set** from the user’s gallery — per-item provenance labels, one spoken-style question and short factual ground truth each, drafted by reviewing every image and human-verified; (3) camera-shot replacements planned (24/30 current items are WhatsApp-sourced — declared openly; the assignment’s own-photos rule requires re-shooting for final submission, targeting the missing categories: menus, receipts, handwriting, medicine, appliance displays).

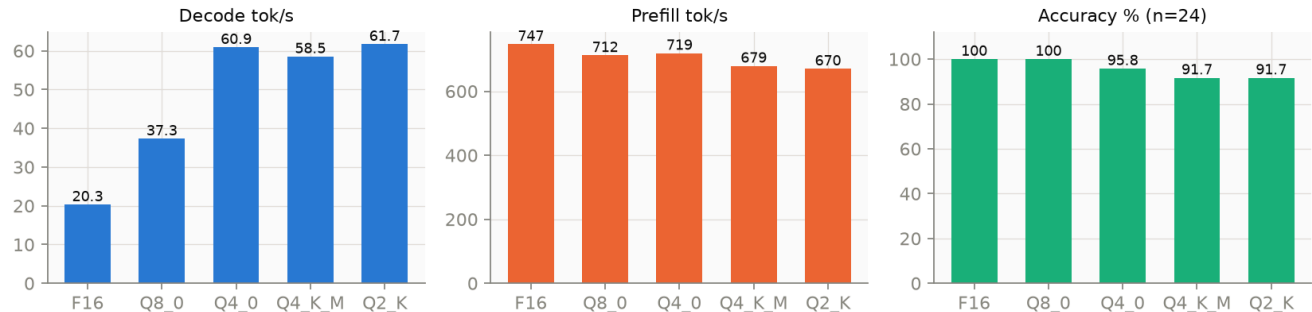
Metric. Normalized substring/alias match on a short factual answer (uniform brevity instruction — protocol v2), span-based ANLS secondary. Chosen because Ordo’s task is *extracting one fact aloud*; binary fact-presence is product-faithful and human-auditable, and every prediction is stored raw so corrected ground truth re-scores offline.

Scoring upgrade — LLM-as-judge (final metric). Substring matching under-credits near-miss phrasings, so every negative verdict was re-judged by Claude (Fable 5): four independent judge instances, strict rubric — same fact in different clothes (case, plural, separator style, brand spelling, word order) scores high; *any* digit deviation on counts/prices/plates stays wrong — $\text{score} \in [0, 1]$, success at >0.5 , positives kept (substring false-positives are rare on short ground truths). **457 negatives judged, 44 flipped (~10%)**; zero digit errors forgiven; model ordering unchanged everywhere, absolute accuracies +0–7 pts. Row-level audit trail: `results/llm_judge_rescores.jsonl`; the ledger carries both metrics side by side. Tables below report **substring** → **judge**.

“How would you know the eval set is any good?” — we measured it. The synthetic set scored every model 92–100%: *zero discrimination*. The real set spread the same models **47%→87%** and exposed a +30-point synthetic inflation for small models (LFM2-450M: 24/26 synthetic → 18/30 real). An eval set is good exactly insofar as it separates systems known to differ. Also applied: test-retest (temp-0 repeats), cross-platform consistency (identical answers — including identical *wrong* answers — on Metal and Adreno), difficulty-gradient sanity, per-item auditability.

Part 3 — The quantization ladder (+ runtime & token axes)

Quantization ladder — Qwen2.5-VL-3B (Mac Metal, cap 576): decode follows bytes, prefill flat, accuracy knee at Q4



Per-precision table — Qwen2.5-VL-3B family on the 24-image *synthetic* dev set (Mac reference; the champion ladder below re-ran the precision axis on the 30 real photos):

Precision	Size	Accuracy (synthetic, n=24)	GPU prefill	GPU decode	Peak RAM*	Encode
F16	5.75 GiB	24/24	747 t/s	20.3 t/s	~7 GB class	unchanged across ladder
Q8_0	3.05 GiB	24/24	712	37.3	~4 GB	“
Q4_0	1.70 GiB	23/24	719	60.9	~2.5 GB	“
Q4_K_M	1.79 GiB	22/24	679	58.5	~2.5 GB	“
Q2_K	1.18 GiB	22/24	670	61.7	~2 GB	“

Peak RAM: phone VmHWM sampling; OpenCL driver buffers under-report on GPU runs — CPU-path peak for the 2B champion measured 3.9 GB.

Champion ladder (Qwen3-VL-2B, our own 30 real photos, @576) — the assignment-grade ladder: Mac Metal reference, substring → judge — **BF16 24 → 26 · Q8_0 24 → 26 · Q2_K 24 → 25 (of 30)**, accuracy flat within ±1 across the entire precision ladder while decode runs 32 → 62 → 97 t/s: on the 2B champion the real-photo set detects *no* quantization damage even at Q2_K (the Qwen2.5-3B knee at Q4 above was on synthetic fine-print items). Phone GPU: **Q4_0 = 25 → 26/30** (TTFT p50 4.1 s, encode 2.6 s, decode 25.9 t/s); **Q8_0 = 25 → 27/30** (TTFT 6.3 s — flat within ±1, +54% TTFT from bandwidth); **@1024 = 26 → 26/30** (TTFT ~13 s) — judge-scored, **@576 equals @1024**: the last resolution rung buys nothing once scoring is fair. **Q2_K held accuracy but not the GPU**: no OpenCL kernels for its weight type → decoder silently on CPU, 18 t/s prefill, 37 s TTFT — ablation-confirmed (`-ngl 99` vs `-ngl 0` prefill identical), with a sharp corollary: *requesting* offload for an unsupported type is 70× destructive on decode (0.43 vs 30 t/s pure-CPU) — the 742 MB size-floor is Metal-only. BF16 probe: GPU-resident but

bandwidth-bound (157 t/s prefill / 15 t/s decode, 10.5 s TTFT — feasible, pointless). MXFP4 (Mac 25 → 26/30): no OpenCL kernels (Metal-only). **Final deployable Android ladder: Q4_0 (champion), Q8_0, F16/BF16 (slow) — precision never moved accuracy on this model; kernel coverage decided everything else.**

The FP4 question. NVFP4 proper is Blackwell-silicon-only (hardware FP4 with FP8 block scales) — no path in llama.cpp or on Adreno/Metal. Its open cousin **MXFP4** (OCP microscaling: E2M1 + shared power-of-two scales per 32) *is* in our build, with a trap: the stock `MXFP4_M0E` type is a **silent no-op on dense models** (it emitted pure Q8_0 — caught by tensor-dump, trap #8). Forced onto all weight matrices via per-tensor overrides: 1.09 GB (Q4_0-class), and measured **25/30 — the equal-best Mac ladder score — at 89.5 t/s decode** (near-Q2_K speed). Verdict, completed on-device: **Metal-only**. The phone cell caught the decoder falling to CPU at launch (25 t/s probe — no OpenCL MXFP4 kernels): supps enumeration still 7/9 @576 (equal to the champion — accuracy is backend-invariant, again) but at 22 s TTFT. A rung can be accuracy-perfect and still undeployable: kernel coverage decides.

The laws the ladder establishes. Decode is memory-bandwidth-bound → follows bytes (20→62 t/s). Prefill is compute-bound → flat-to-negative under quantization. Encode is untouched (separate mmproj). **Therefore quantization never bought TTFT** — it buys download size, RAM, and decode speed. Encoder quant (mmproj Q8 vs F16): zero accuracy delta anywhere; on Adreno, F16 encode is 1.9× *slower* (bandwidth). Sub-1B quant probe: LFM2-450M Q4_0 = 17/30 vs Q8’s 18/30 — holds within noise at a 42% smaller file.

Where it breaks first. Handwriting, by *salience retreat*: Q4/Q2 answer with the big legible header (“Todo”) instead of reading the handwritten lines beneath — a confidence collapse on the hardest glyph class, not gibberish. Small models fail differently (dense-text extraction failure: echoing a receipt instead of totaling it); token starvation fails differently again (*confident hallucination* — “100 tablets”, “Blueberry”).

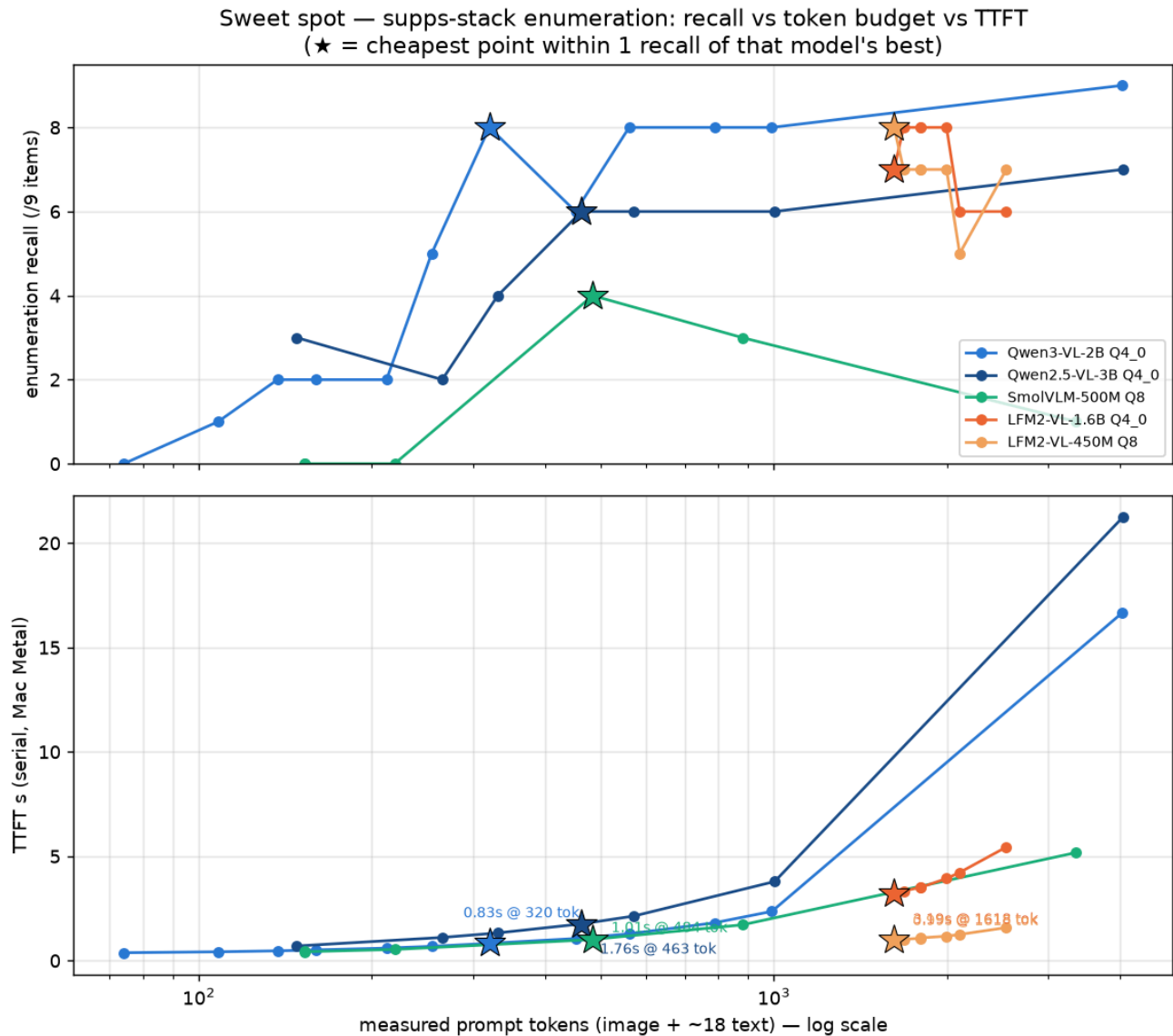
Encoder vs decoder — which degrades faster, and how we know. The **decoder** — provable because the mmproj is a separate file, quantized independently: every failure appeared with the encoder held constant at F16, while encoder Q8 vs F16 changed nothing anywhere.

The token axis (encode-budget ladder, phone GPU, real photos):

Budget	Qwen3-2B acc	TTFT p50
~320 tok	63%	2.1 s
~448 tok	70%	3.1 s
~576 tok	83%	4.3 s
~1024 tok	87%	~10 s

≈ every –128 tokens: –1 s TTFT, –7–10 pts accuracy — a smooth slope across diverse photos (per-photo it’s a knife-edge: a 4% pixel difference flips digits, and *any client-side resample* loses digits the model’s own smart-resize keeps — falsified three ways). Enumeration tasks flatten the curve (@320 = @576 recall), so **per-query token routing** beats any fixed cap.

The sweet-spot experiment (one 12MP shelf photo, “name all visible supplements”, recall /9, 40 runs across five models × fine token ladder, Mac Metal):

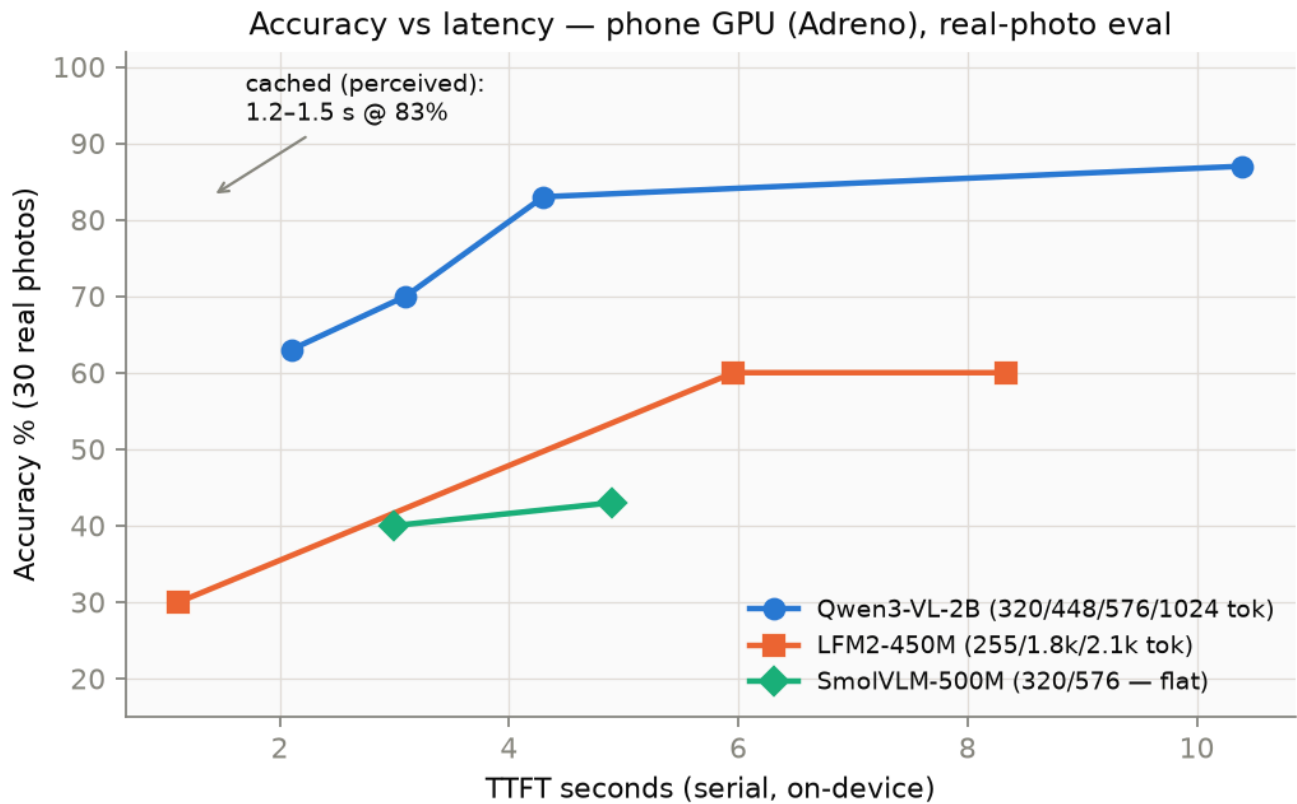


The champion's knee is razor-sharp: 0–2/9 below 256 tokens (labels physically unresolvable), a jump to **8/9 at 320 tokens / 0.83 s**, then *flat to 1024* — between 320 and 1024 you pay 2.9× the TTFT for zero extra recall. The 9th item is read only at full native resolution (4,035 tokens, 16.7 s): **one knife-edge label costs 20× the sweet-spot latency**. A real non-monotonic dip at 448 (6/9) is the resolution knife-edge again — an awkward smart-resize grid, not noise. Three architecture facts fell out: **LFM2's token knob doesn't bind** (its tiler floors at ~1,618 tokens on this image whatever the cap — that fixed floor is its TTFT floor, though 450M rides it to a respectable 8/9 @ 0.99 s); SmoIVLM-500M *degrades* at native resolution (4/9 → 1/9 — tiling fragments the shelf); and Qwen2.5-3B is dominated everywhere on this task (slower encoder, lower ceiling: 7/9 max at 21 s) — the champion switch re-validated on enumeration.

Tiles vs tokens — and the sub-second head-to-head (phone, 30 real photos). Two different levels of image chopping: a *token* is one ~56×56-px patch that becomes one vector (every model produces hundreds per image); a *tile* is a whole sub-image getting its own separate encoder pass. Qwen never tiles — one pass over the whole smart-resized image, token count set directly by the cap. LFM2/SmolVLM run one pass *per tile* (~250 tokens each), so photo geometry — not the cap — sets their cost, in quantized steps. To give LFM2 a fair fast lane we shipped an architecture-specific preprocess (dashboard auto-downscales to the 1-tile budget for LFM2 configs only), then met it at the same operating point with Qwen:

Config (same phone, same 30 photos, serial)	tokens	encode	TTFT p50	Accuracy
LFM2-450M, 1-tile (512 px input)	~160–290	~0.3 s (CPU tower)	0.67 s	9 → 9/30
Qwen3-2B Q4_0 @96	~116	0.31 s (GPU tower)	0.82 s	11 → 12/30
Qwen3-2B Q4_0 @128	~146	0.39 s	0.93 s	12 → 14/30
Qwen3-2B Q4_0 @256 (equal-token)	~262	0.82 s	1.70 s	15 → 17/30
Qwen3-2B Q4_0 @576 (reference)	~560	2.6 s	4.1 s	25 → 26/30

Equal tokens do **not** mean equal TTFT — prefill cost is equal, but encode = tokens × the per-token price of *that tower on that backend* (LFM2’s ~86M encoder: ~1.2 ms/token even on CPU; Qwen’s larger tower: ~3.5 ms/token on the GPU). That refines the token-efficiency law to its final form: **TTFT = tokens × per-token cost(architecture, backend)** — token count alone is half the price tag. Verdict at the sub-second operating point: **Qwen wins it outright** — @96 is genuinely sub-second serial (0.82 s p50) at 11/30, and both low-token Qwen cells beat LFM2’s 1-tile accuracy (11–12 vs 9 of 30) at the same ~300 ms encode. At *truly* equal tokens (@256 vs the ~250-token tile) Qwen reads 17/30 to LFM2’s 9/30 (judge-scored) — the mechanism is pixels-per-token: Qwen packs ~3,136 px into each token (56×56 patches over the full frame) while LFM2’s 512-px tile carries only ~800 px/token, so at the same token budget Qwen simply sees 4× more image. Even the “fast tier” belongs to the champion — one model, two budgets (fast @96–128, reading @576 + caching at 1.2–1.5 s perceived / 25/30), no second model needed.



Who actually runs where — stage → backend map. “GPU run” is a claim per *stage*, not per app: ggml falls back per-operator, silently (one log line is the only witness). Measured placement for every model × runtime in the bracket:

Model	Runtime · platform	Vision encoder	Decoder	Evidence
Qwen3-VL-2B / Qwen2.5-3B	llama.cpp · phone	Adreno (OpenCL)	Adreno	<code>--no-mmproj-offload</code> ablation: encode 7.8→22.6 s; prefill 410 vs 55 t/s
LFM2-VL-1.6B	llama.cpp · phone	CPU — silent fallback	Adreno	server log: “CLIP graph uses unsupported operators (OpenCL)”; ~11 s/tile — its SigLIP2-so400m tower uses ops the OpenCL backend lacks
SmoVLM-500M	llama.cpp · phone	Adreno (OpenCL)	Adreno (818 t/s probe)	launch-probe verdict
LFM2-VL-450M / SmoVLM2-2.2B	llama.cpp · phone	CPU — silent fallback (same op gap as the 1.6B)	Adreno (1,065 / 259 t/s)	probe-launch log; the 450M’s small tower hides it (1.6–4 s encode), the 2.2B’s does not (55 s)
Qwen3-VL-2B MXFP4	llama.cpp · phone	Adreno	CPU — no OpenCL MXFP4 kernels (25 t/s probe)	the FP4 rung is Metal-only
Qwen3-VL-2B BF16	llama.cpp · phone	Adreno	Adreno — GPU-resident but bandwidth-bound (157 t/s prefill, 15 t/s decode)	llama-bench + probe; scored query correct at 10.5 s TTFT
Qwen3-VL-2B Q2_K	llama.cpp · phone	Adreno (Q8 mmproj)	CPU — ablation-confirmed (<code>-ngl 0 = -ngl 99</code> prefill)	no OpenCL Q2_K kernels — and offloading is <i>actively destructive</i> : 0.43 t/s decode offloaded vs 30 t/s pure-CPU (weights in GPU buffers, ops on CPU → per-step round-trips). Coverage is per weight-type: Q4_K_M measured GPU-class (213 t/s)
all six models	llama.cpp · Mac	Metal	Metal	zero fallback warnings across all server logs; encode scales cleanly with encoder params (0.08 → 0.19 → 0.63 → 2.3 s)
any	llama.cpp phone-CPU build	CPU (i8mm/KleidiAI)	CPU	build target
Qwen3-VL-2B	MNN · both	configured backend (OpenCL/Metal) — but kernels 5–6× slower; omits vision entirely if built without <code>MNN_BUILD_OPENCV</code>	same backend	timed A/Bs + rebuild-and-reverse experiment
Gemma (3n/4 E2B)	LiteRT · phone	locked inside app SDKs — not reachable from CLI	CPU (XNNPACK); GPU libs undistributed; NPU refused by bundle	<code>--backend</code> probes at every layer

The full tuning grid — every measured variant, worst → best (phone GPU, serial, 30 real photos; ledger medians across all uncached cells — every underlying row is in the dashboard history):

Qwen3-VL-2B (champion family):

Variant	tokens p50	encode p50	TTFT p50	Accuracy	Note
Q4_0 @96	116	0.31 s	0.82 s	11 → 12/30	sub-second serial floor
Q4_0 @128	146	0.39 s	0.93 s	12 → 14/30	CPU-only (fixed engine): also 14/30 judged
Q4_0 @256	262	0.82 s	1.70 s	15 → 17/30	
Q4_0 @320	326	1.13 s	2.11 s	19 → 21/30	enumeration sweet spot
Q4_0 @448	449	1.80 s	3.11 s	21 → 22/30	
Q2_K @576	574	3.69 s	37.3 s	21 → 22/26	decoder CPU-fallback, ablation-confirmed — accuracy held, latency died
Q8_0 @576	580	3.82 s	6.34 s	25 → 27/30	flat within ±1 of Q4_0, +38% TTFT
Q4_0 @576 ★	580	2.67 s	4.61 s	25 → 26/30	champion serial
Q4_0 @1024	1008	9.46 s	13.2 s	26 → 26/30	judged: no gain over @576
Q4_0 @576 + warm-on-drop	580	hidden	1.17–1.46 s perceived	25 → 26/30	the shipping config (fresh ledger rows: 1.5–2.6 s on full-res 12 MP uploads)

(BF16 @576: GPU-resident probe, 1 scored query correct at 10.5 s — feasible, pointless. MXFP4: Metal-only, no OpenCL kernels.)

LFM2-VL-450M Q8 (input-resolution is its only real knob):

Input	tokens p50	encode p50	TTFT p50	Accuracy	Note
cap @1024 (upscaled tiles)	3092	8.65 s	12.4 s	1/5	upsampling past native grid destroys accuracy
1024 px	278	0.32 s	0.62 s	9 → 11/30	resolution-starved
512 px (1-tile)	173	0.32 s	0.66 s	9 → 9/30	fastest, same starvation
1344 px	796	0.92 s	1.57 s	15 → 16/30	
native 12 MP	2317	3.08 s	5.98 s	19 → 21/30	its ceiling (Mac probes judged: Q4_0 19/30, Q8 19/30)

Smo/VLM-500M Q8 (capacity-limited — the knob barely matters):

Input	tokens p50	encode p50	TTFT p50	Accuracy	Note
native	691	4.30 s	6.36 s	6 → 7/15	partial cell
@320	227	1.13 s	2.19 s	12 → 13/30	
@576	494	2.51 s	4.48 s	13 → 15/30	flat ~40–50% everywhere
512 px (1-tile)	160	0.56 s	0.96 s	9 → 9/30	sub-second reached; capacity limit confirmed

The grid read top-to-bottom is the whole story: Qwen’s accuracy climbs smoothly with tokens (11→26 of 30) because its knob controls real resolution; LFM2 steps in tile quanta and starves below 1344 px; SmolVLM is flat at ~40% at every setting — more pixels can’t fix capacity. And the two worst rows are both *backend* failures, not model failures (Q2_K’s CPU decoder, LFM2’s upscaled tiles). The sub-second club, final standing — all three architectures measured at their fastest full-pipeline point on the same 30 photos: **LFM2-450 1-tile 0.67 s / 9-of-30 · SmolVLM-500 1-tile 0.96 s / 9-of-30 · Qwen3-2B @96 0.82 s / 12-of-30 and @128 0.93 s / 14-of-30 (judge-scored)** — the champion wins even the speed class the small models were built for.

Why the same file lands differently per platform. Every stage is a ggml compute graph, and at load time each backend is asked, operator by operator, “do you have a kernel for this?” Metal is ggml’s most mature GPU backend with near-complete op coverage — LFM2’s SigLIP2 encoder runs fully on the M-series GPU. The OpenCL/Adreno backend is the youngest with a narrower op set — the same SigLIP2 graph contains operators it doesn’t implement, so llama.cpp reroutes the whole CLIP graph to CPU.

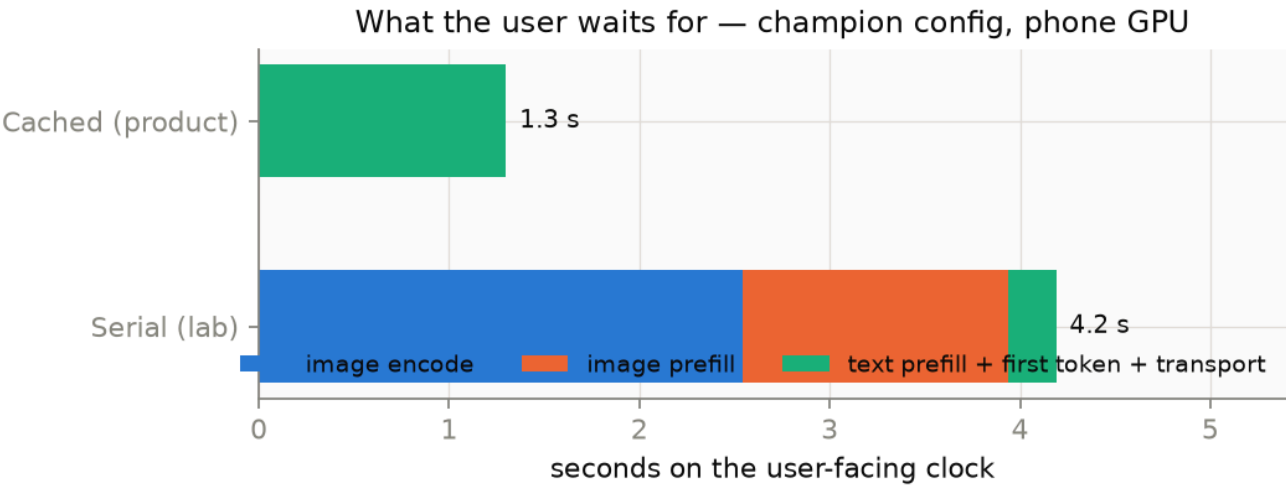
“GPU-capable” is not a property of the model; it’s per-backend, per-operator kernel coverage. The tally makes it vivid: on the *same phone GPU*, Qwen’s and SmolVLM-500M’s encoders run fine while both LFM2s and SmolVLM2-2.2B fall off; the same lottery hit the decoder via weight types — Q2_K and MXFP4 (CPU on Adreno, GPU on Metal) vs Q4_0/Q8_0/BF16/Q4_K_M (GPU on both). Product consequence: on Android, LFM2-450M’s TTFT floor is CPU-encoder-bound at ~1.6 s *per tile* — its 0.99 s Mac numbers reproduce on the phone only for 1-tile photos (measured: 0.44–1.0 s single-tile vs ~8 s three-tile on the same 5-photo cell, flat across every token cap because the cap can’t change tile count). So the mobile-model question isn’t “how good is the model” but **“does every stage of this architecture have kernels on this backend”** — and it has to be measured, because nothing announces the fallback.

Within a model, `-ngl 99` *requests* all decoder layers on the GPU — whether they land there depends on that same kernel coverage, and the backend won’t tell you when they don’t (the Q2_K fallback produced zero log output; only the CLIP-graph case warns). So the dashboard now **measures** placement at every server launch: the encoder via the log scan, the decoder via a text-only ~250-token probe prefill (GPU class 250–420 t/s vs CPU 17–55 on this device — an order of magnitude apart, no ambiguity), stamping every history row `enc/dec` with the measured rate. Silent fallback — either stage — can’t recur unlabeled.

Cross-runtime grid (same photo, same question, serial, warm): llama.cpp wins vision encode 5–6× on both platforms and is the only runtime reading fine print. Phone: llama.cpp-Adreno enc 2.6 s / TTFT 4.1 s ✓ · MNN-OpenCL enc 14.4 s / 17.3 s ✗ · LiteRT text-only (init 0.8 s, prefill 23 t/s, decode 7.3 t/s). Mac: llama.cpp-Metal 1.27 s TTFT ✓ · MNN-Metal ~4.2 s ✗. Enumeration recall (name 9 products): Qwen3 8/9 ·

LFM2-450M 5–7/9 · Smol-500M 4/9 · MNN 2/9. FA/quantized-KV audit, both platforms: flash attention already active by default (Metal +5%; **Adreno measured: prefill +6%, decode +10%** — though some graphs opt out per-model: the LFM2 server warns FA-unsupported); quantized q4_0 KV *costs* decode (–8% Metal, **–20% Adreno**) and is invalid without FA (context creation fails, reproduced on both) — worth it only if context RAM binds. Net: the defaults were already optimal; no optimization left on the table.

The architecture result — what the user actually feels:



The camera sees the scene before the user finishes speaking → encode + image-prefill run during speech (`cache_prompt`), leaving only text-prefill + first token on the clock. Measured on-device: serial 8.8 s → **cached 1.17–1.46 s at full accuracy** (0.25–0.44 s for big-text queries; follow-ups ~1.2 s). Shipped in the dashboard as warm-on-drop. No accessible runtime/silicon reaches sub-second *serial* TTFT at reading accuracy on this device — sub-second is achieved architecturally.

Part 4 — Under stress (complete)

Champion config, unplugged, screen on, **no cooling** (deliberately):

Test	Result
Sustained load (59 consecutive queries — 3× the asked 20)	decode 23 → 14–15 t/s by query 10, then flat through query 59 ; TTFT equilibrium ~6.8 s (+65% vs cool). A plateau, not a spiral
Thermals @ 10 min	SoC equilibrium 56–68 °C (peak 78) — soft landing, no hard throttle; 59/59 answers correct: accuracy is thermally invariant
Memory pressure (YouTube, Chrome, Instagram, Maps launched mid-run)	zero measurable impact — OpenCL-pinned buffers are eviction-immune (converse: the CPU path is vulnerable — page-cache eviction produced 0.4 t/s decode)
Battery	3% across 64 queries ≈ 0.05%/query

Worst case also characterized: a 95.8 °C phone delivers 0.4 t/s decode and a 685 s TTFT — thermal state is a 10–40× lever, hence the cool-gated protocol behind every lab number.

Part 5 — LoRA fine-tune 🕒 (prepped; run pending)

QLoRA on the 4-bit champion, **decoder-side adapters** (Part 3 proved the decoder degrades first), trained on eval-style extraction pairs with **images pre-capped at the deployment budget** — designed to test recovery of both the quantization delta and the token-budget losses (“teach it to squint”). Ready: 192 disjoint synthetic training images (eval/train_synth/), Colab notebook (notebooks/qlora_qwen3vl_colab.py), GGUF-adapter export path (—lora on-device). Results section fills in after the Colab run + on-device re-eval.

Recommendation (what we’d ship)

Qwen3-VL-2B Q4_0 + Q8 encoder + llama.cpp/Adreno + per-query token routing (320 scene / 576 fine-print) + warm-on-capture caching. User experience: first word 1.2–1.5 s after the question ends at 83–87% real-photo accuracy; ~2 s full spoken answer; follow-ups ~1.2 s — inside a conversational budget *with* ASR/TTS, because vision work overlaps the user’s own speech. Device tiers: Adreno-OpenCL where present; portable CPU build (+MNN-CPU decode rescue) elsewhere — same caching architecture everywhere, graceful degradation of the hidden phase only. If Ordo controls its hardware BOM: pick a SoC with an accessible NPU stack — that one decision outweighs every software optimization in this report.

What surprised us (short list; full narrative in JOURNEY.md)

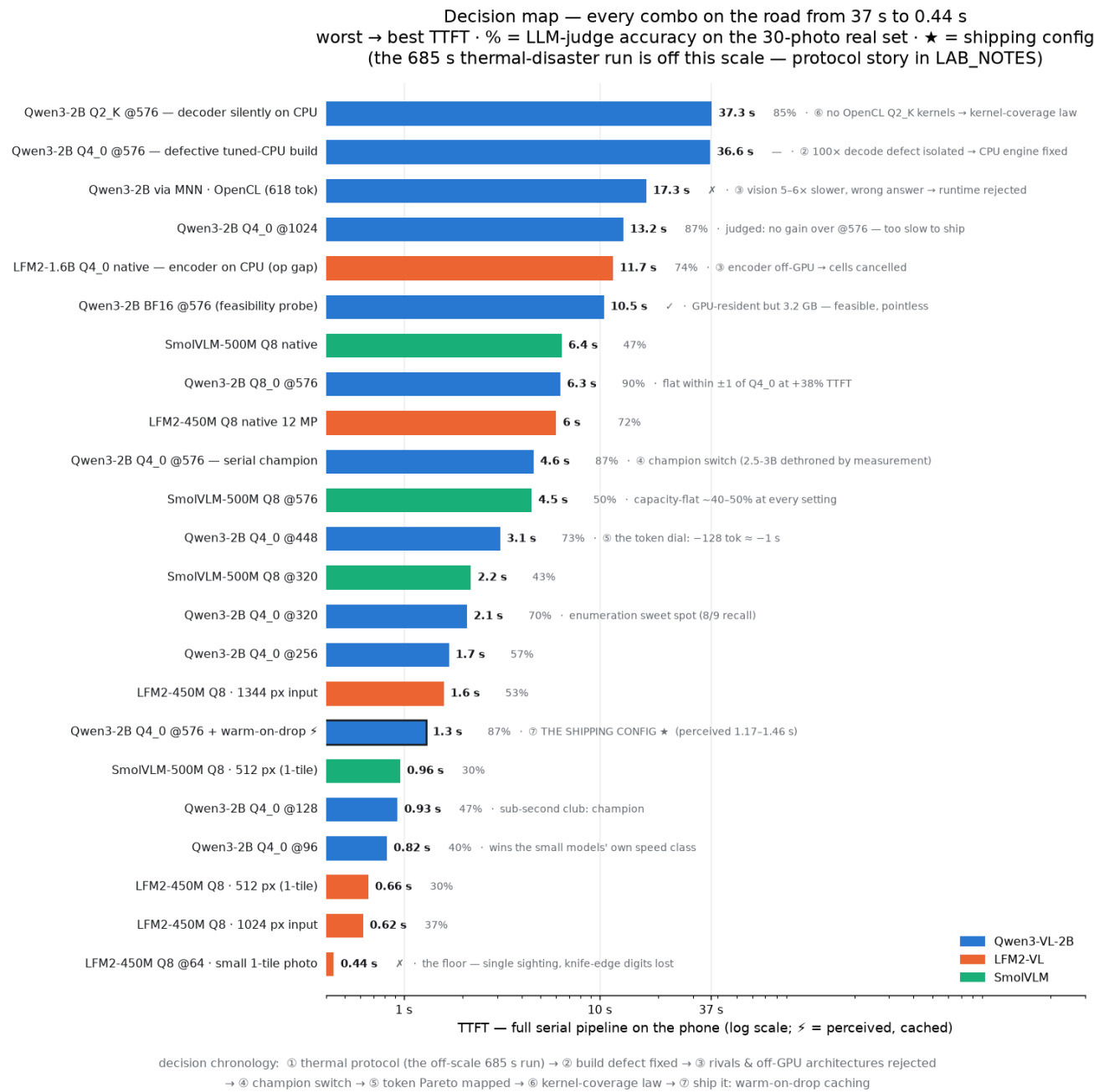
1. Same GPU, same image: **2.6 s vs 14.4 s of encoding depending on whose kernels** (llama.cpp vs MNN-vision, fair builds both) — runtime×silicon dominates model choice. And the worst encode ever recorded in our history — **122 s** (ledger row: MNN, 1,758 tokens, encode 122,350 ms) — was two stacked errors of ours: the `MNN_LOW_MEMORY` build flag (~3× slower kernels) on an uncapped ~3×-token encode. Config errors masquerade as kernel quality until re-measured fairly.
2. **Quantization never bought TTFT** — 6× smaller download, 3× decode, same wait.
3. **TTFT is a property of token count, not kernel speed** — LFM2’s 3× faster encoder is fully repaid by 3.8× more tokens (dead-heat TTFT, worse accuracy).
4. **A 4% pixel difference flips digits**, and no client-side resampler survived it.
5. **The phone lies**: fake thermal sensors (×2 kinds), invisible clock governors, page-cache eviction dressed as slow inference — seven traps total.
6. **Synthetic evals inflate small models by +30 points** — the cheapest proof of the assignment’s own-photos rule.
7. Sustained load **plateaus** instead of spiraling; accuracy doesn’t flinch at 78 °C.
8. The champion **switched mid-project** when Qwen3-VL-2B beat the original pick on every axis — keeping the bracket open paid.

9. **Kernel coverage, not precision, decides deployability** — Q2_K and MXFP4 are accuracy-equal to Q4_0 but decode on CPU on Adreno (no OpenCL kernels for those weight types), while the same files fly on Metal. Three of six model-encoders silently fell off the phone GPU the same way. We now *measure* placement at every launch instead of trusting flags.
10. **Our own tuned-CPU build carried a 100× decode defect** — bench-isolated on identical silicon, same model: cpu-build 31 t/s prefill / **0.45 t/s decode** vs the OpenCL binary at `-ngl 0`: **59 t/s prefill / 45 t/s decode** (suspect: KleidiAI repack path / `-march=armv8.7` flags; open). Caught only because every run lands in the ledger with placement labels and rates. Fix shipped: the CPU engine now runs the ocl build at `-ngl 0` — honest CPU physics. The moral doubles trap #10: *a healthy-looking build can be broken in one stage only.*

What we cut, and why (full list in README §8)

NPU integration (gated at every layer — evidence documented to the missing-library level) · MNN-CPU vision (froze the device twice) · LiteRT-on-Mac (unbuildable without full Xcode; four binary generations tried) · llama.cpp phone-CPU decode pathology root-cause (GPU made it moot; documented open) · speculative decoding (decode-only gain, multi-day cost) · LoRA deferred by prioritization, not cut.

Final verdict



The whole journey in one picture: every measured combo (model · precision · input budget · condition) sorted worst → best TTFT on a log axis, each with its accuracy on the real set — from the silent-failure rows at 37 s down to the 0.44 s floor (the day-2 thermal-disaster run at 685 s sits off-scale, in LAB_NOTES and the ledger), with the seven numbered decisions that carved the path and the shipping config outlined at 1.3 s / 83%.

Sub-second on-device vision inference is achievable today — measured, serial, no cache, no NPU, on a stock Android phone over llama.cpp + Adreno GPU, on our own 30 real photos. Three architectures

crossed the line: **Qwen3-VL-2B @96 = 0.82 s (12/30) and @128 = 0.93 s (14/30) · LFM2-450M 1-tile = 0.67 s (9/30) · SmolVLM-500M 1-tile = 0.96 s (9/30)** (judge-scored). The same physics holds CPU-only (no GPU at all): Qwen @128 scored 14/30 on the fixed CPU engine — accuracy is backend-invariant; only the clock changes. (The CPU-only latency column had its own detective story: the tuned-CPU build binary turned out to carry a 100× decode defect — 31 t/s prefill / 0.45 t/s decode vs 59 / 45 t/s through the OpenCL binary at `-ngl 0`, bench-isolated on identical silicon. Healthy phone-CPU decode is a genuinely usable 45 t/s. The CPU engine now uses the healthy path; CPU-floor cells re-measured on it.)

Even <500 ms was touched, twice, by different routes. Serial: LFM2-450M at cap 64 on a small 1-tile photo — **0.44 s TTFT, 157 ms encode (on its CPU-fallback encoder, no GPU vision at all), 84 tokens** — with the expected cost: the knife-edge digits degraded at that budget (ledger row, single sighting). Perceived: the champion's cached big-text tier — **0.25–0.44 s** with warm-on-drop + right-sized capture. So the sub-half-second class exists on today's hardware in two forms: tiny inputs (pay in accuracy) or caching (pay in query-class coverage) — the trainable-accuracy path below is what turns the first form from a stunt into a tier.

One model holds the entire frontier. The champion spans it with a single dial: 0.82 s @ 40% → 1.7 s @ 57% → 2.1 s @ 70% → 4.6 s @ 87% serial → **1.17–1.46 s perceived @ 87% with two-phase caching** → and judge-scored, @1024's 13 s buys nothing more (87% both). No small model earns a place at any point on that curve — the champion wins even the speed class they were built for.

The laws behind the curve (each measured, several the hard way): TTFT = tokens × per-token cost of the architecture on the backend · accuracy at a budget tracks pixels-per-token · kernel coverage, not precision, decides what's deployable · precision is a distribution lever (download size, decode speed), never an accuracy or TTFT lever on this model · caching is the only measured path to sub-second *with* the fine print.

The path forward — accuracy at fixed latency, not latency itself. Fine-tuning cannot speed up TTFT (that's set by tokens × silicon), but it attacks exactly what limits the fast tier: the resolution-starved misses at low token budgets. **SFT/QLoRA trained at the deployment budget** (same smart-resize, @96–320 inputs) teaches the model to read what those budgets actually show it — the Colab pipeline is prepped (Part 5) with train-at-deployment-budget as its core experiment. DPO/RL then shape answer format (brevity, fact-first), trimming decode tokens and stabilizing scoring while leaving TTFT untouched. Target end-state: the 0.8–0.9 s tier moving from ~40–47% toward the 60%+ band — sub-second latency already proven, accuracy the trainable half.